The horizontal distance travelled by the cricket ball

$$R = \frac{(30)^2 \times \sin 60°}{10} = \frac{900 \times \frac{\sqrt{3}}{2}}{10} = 77.94\,m$$

This distance is greater than the distance of the boundary line. Hence the ball will cross this line and go for a six.

## 2.11.4 Introduction to Degrees and Radians

In measuring angles, there are several possible units used, but the most common units are degrees and radians. Radians are used in measuring area, volume, and circumference of circles and surface area of spheres.

**Radian** describes the planar angle subtended by a circular arc at the centre of a circle. It is defined as the length of the arc divided by the radius of the arc. One radian is the angle subtended at the centre of a circle by an arc that is equal in length to the radius of the circle. This is shown in the Figure 2.44.

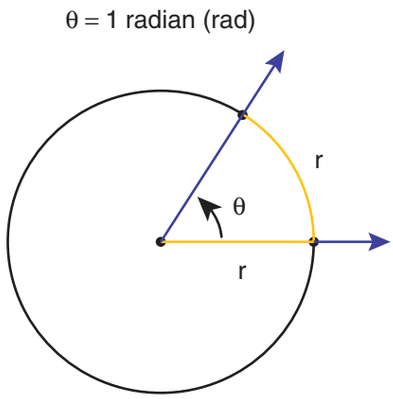



**Figure 2.44** One radian (shown in yellow color)

Degree is the unit of measurement which is used to determine the size of an angle. When an angle goes all the way around in a circle, the total angle covered is equivalent to 360°. Thus, a circle has 360°. In terms of radians, the full circle has $2\pi$ radian.

Hence we write $360° = 2\pi$ *radians*

$$\text{or } 1 \text{ radians} = \frac{180}{\pi} \text{ degrees}$$

which means $1 \text{ rad} \cong 57.27°$

### EXAMPLE 2.39

Calculate the angle $\theta$ subtended by the two adjacent wooden spokes of a bullock cart wheel is shown in the figure. Express the angle in both radian and degree.



### *Solution*

The full wheel subtends $2\pi$ radians at the centre of the wheel. The wheel is divided into 12 parts (arcs).

So one part subtends an angle $\theta = \frac{2\pi}{12} = \frac{\pi}{6}$ radian at the centre

Since, $\pi$ *rad* $= 180°$, $\frac{\pi}{6}$ radian is equal to 30 degree.

∴ The angle subtended by two adjacent wooden spokes is 30 degree at the centre.



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